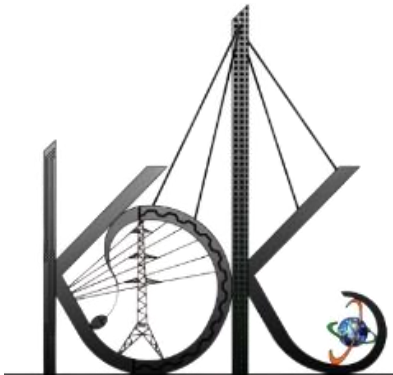


SUMMER INTERNSHIP PROJECT REPORT
ON
SUPPLY CHAIN MANAGEMENT ANALYTICS USING
POWER BI

A Report submitted to R.T.M NAGPUR University in partial fulfilment of course
MASTER OF BUSINESS ADMINISTRATION (MBA)

SUBMITTED BY
(Ms Vaishnavi Amit Ojha)

GUIDED BY
(Prof. Deepa P. Bagde)



DEPARTMENT OF MANAGEMENT STUDIES
K. D. K. COLLEGE OF ENGINEERING, NAGPUR.
(2025-2026)

KARMAVIR DADASAHEB KANNAMWAR
COLLEGE OF ENGINEERING, NAGPUR
DEPARTMENT OF MANAGEMENT STUDIES
SESSION 2025-2026

CERTIFICATE

This is to certify that the internship report titled ‘**SUPPLY CHAIN MANAGEMENT ANALYTICS USING POWER BI**’ has been carried out by **Ms. Vaishnavi Ojha** during the Summer Internship Project **2025-26**. The study was done in the organisation, LiveTechSkills and is approved.

Prof. Deepa P. Bagde
Guide

Prof. R. S. Budhe
HOD, MBA
KDKCE

Dr. V. P. Varghese
Principal
KDKCE

INTERNSHIP CERTIFICATE

LIVETECHSKILLS

13, 1st Floor, Block 2, MB Tower, Mire layout,
Jawahar Nagar, Nandanvan Main Road, Nagpur. 440009
Contact: 7507111167/9096060149

LiveTechSkills

Date: 26/08/2025

INTERNSHIP COMPLETION LETTER

This is to certify that **Ms. Vaishnavi Ojha** has successfully completed her internship with **Livetechnskills** as a **Business Analyst Intern**.

Her internship tenure was from **4th July 2025 to 25th August 2025**. During this period, she was actively involved in analyzing business requirements, preparing documentation, supporting project workflows, and contributing to process improvements under the guidance of our team.

We commend your contributions to tasks and projects, where you displayed creativity, problem-solving skills, and attention to detail. Your positive attitude, willingness to learn, and collaborative approach were instrumental in achieving project milestones and delivering high-quality results.

We found her to be sincere, diligent, and enthusiastic in carrying out the assigned tasks and responsibilities. Her contributions added value to the projects she was associated with.

We wish her all the best in her future career endeavors.

Sincerely,
Mr Akshay S. Jain
Centre Head,
LIVETECHSKILLS

Mo: [9096060149](tel:9096060149) | www.livetechnskills.com


FOR LIVETECHSKILLS
PARTNER

LiveTechSkills

ACKNOWLEDGEMENT

With Profound feelings of immense gratitude and affection, I would like to thank my guide **Prof Deepa P. Bagde**, co-ordinate for her continuous support, motivation, enthusiasm and guidance. Her encouragement, supervision with constructive criticism and confidence enable me to complete this report.

I express my admiration for **Dr. A. M. Badar, Vice Principal** for his valuable advice and support throughout this venture.

I also put forth my deepest sense of gratitude toward **Dr. V. P. Varghese, Principal** for constant motivation and providing necessary infrastructure.

Finally, a special thanks to all the faculty members of the department for their cooperation throughout the project work.

Vaishnavi Ojha

Projectee

INDEX

PART	CONTENT	PAGE NO.
1 (A)	Company Profile	6-11
(B)	Sector Overview	12-14
	Project Work	15-19
2 (A)	Actual Work Done	
	Week wise work allocated and work done	20-24
(B)	Findings, Suggestions	25-27
(C)	Conclusions	
	Conclusion	28-30
	Bibliography	31
	Performance Letter	32



1. (A) Company Profile (LiveTechSkills)

Name of the company:	LiveTechSkills)
Sector:	IT Services
Address of Company:	Nagpur, Maharashtra
Date of Incorporation:	Year of Establishment 2014
Register city:	Nagpur
Register State:	Maharashtra
Status:	Active
Website:	https://www.livetechskills.com/services

1. History & Establishment

LiveTechSkills was established in **2014 in Nagpur, Maharashtra** with the vision of bridging the gap between industry requirements and academic learning. The founders, a group of IT professionals and educators, noticed that while universities provided theoretical knowledge, students lacked exposure to real-world projects, industry tools, and employable skills. LiveTechSkills began as a small training institute offering software and IT courses. Within a few years, it evolved into a service-based IT company specializing in training, consulting, and analytics services.

Initially, the company catered to fresh graduates and engineering students, focusing on basic programming and database courses. However, as the IT landscape evolved, LiveTechSkills diversified into Data Science, Business Analytics, Cybersecurity, and Cloud Computing. The early years were characterized by bootstrapped operations, small classrooms, and personalized training, which helped the organization establish credibility and trust among students. Over time, the firm built long-term partnerships with corporates, SMEs, and academic institutions.

Today, LiveTechSkills is not only a training provider but also a solution partner for businesses in domains like data-driven decision-making, cyber defense, and supply chain analytics. The journey reflects an organization that started with a small vision but continuously scaled by adapting to market demands and technology shifts.

2. Services Offered

LiveTechSkills operates as a multi-service IT firm offering a diverse portfolio:

- **Web Design & Development** – Creating modern, responsive websites for businesses and institutions.
- **Cybersecurity Solutions** – Offering consulting, penetration testing, and training to ensure businesses remain secure against modern threats.
- **Data Analytics** – Providing business intelligence dashboards (Power BI, Tableau) and advanced analytics to help companies understand customer behavior, operations, and performance metrics.
- **Data Science** – Delivering machine learning, artificial intelligence, and predictive analytics solutions to enterprises, along with specialized training programs.
- **Business Intelligence (BI)** – Designing and deploying BI solutions to improve decision-making in industries like manufacturing, healthcare, retail, and supply chain.
- **Corporate Training** – Tailored programs for IT employees to upskill in new-age technologies such as Python, R, SQL, AWS, and cloud analytics.

The blend of services enables LiveTechSkills to serve both individuals (students, professionals) and organizations (corporates, SMEs), positioning itself as a holistic service provider in the IT sector.

3. Client Base & Market Presence

Over the last decade, LiveTechSkills has built a strong regional and semi-national presence. Its primary market is Central India, with Nagpur as its headquarters, but it also caters to clients in Mumbai, Pune, Hyderabad, and Bangalore. The company has successfully collaborated with 50+ SMEs and start-ups in domains like logistics, healthcare, retail, and education, providing analytics-driven solutions.

In the training sector, LiveTechSkills has partnered with universities and engineering colleges to deliver workshops, internships, and campus-to-corporate programs. With 1000+ alumni placed in multinational companies like **TCS, Infosys, Wipro, and Accenture**, the organization has established a strong employer brand. Its alumni network also contributes back by mentoring students and providing placement referrals.

The company's corporate contracts include short-term consulting for supply chain analytics, predictive modeling for sales forecasting, and cybersecurity audits. While its strongest presence remains in Maharashtra and neighboring states, the brand is gradually expanding through online delivery of courses and consulting services across India.

4. Training Methodologies – Agile vs. Waterfall

One of LiveTechSkills defining strengths is its dual methodology approach in service delivery.

- **Agile Methodology** is adopted for analytics projects and BI dashboard development. Here, projects are broken into sprints, allowing iterative improvements and continuous client feedback. For example, in the Supply Chain Management (SCM) Power BI project, dashboards were developed sprint by sprint, with weekly reviews to refine KPIs like Inventory Turnover and Vendor Lead Time. Agile ensures flexibility, adaptability, and alignment with client expectations.
- **Waterfall Methodology** is applied for structured training programs and phased projects. Each training program follows a defined structure: requirement gathering, content design, delivery, assessment, and placement. This ensures consistency in delivery and high-quality outcomes for students.

This combination of Agile (for dynamic projects) and Waterfall (for structured programs) allows LiveTechSkills to balance innovation with reliability, making it effective in both corporate consulting and academic training.

5. Organizational Structure

LiveTechSkills follows a functional organizational structure with a flat hierarchy for innovation:

- **Chief Executive Officer (CEO)** – Provides vision, strategy, and oversees operations.
- **Department Heads** –
 - ✓ Training Head (handles trainers, curriculum, delivery).
 - ✓ Sales & Business Development Head (client contracts, partnerships).
 - ✓ Operations Head (infrastructure, logistics, IT systems).
- **Advisory Council** – Senior industry professionals who guide curriculum updates and technology adoption.
- **Trainers & Mentors** – Certified professionals in domains like Python, Power BI, Cybersecurity, and Data Science.
- **Placement Cell** – A dedicated unit ensuring students' employability through industry tie-ups.

This structure ensures that while decision-making is centralized at the top, execution is distributed among domain leads, making the company efficient and adaptable.

6. Mission, Vision, and Core Values

- **Mission:** To bridge the gap between academic learning and industry requirements by providing cutting-edge training and services in IT, data analytics, and emerging technologies.
- **Vision:** To become a leading national-level IT consulting and training firm recognized for innovation, quality, and industry relevance.
- **Core Values:**
 - ✓ Innovation: Adopting the latest technologies and methodologies.
 - ✓ Integrity: Transparent operations and honest guidance.
 - ✓ Excellence: Commitment to quality in every service.
 - ✓ Student Success: Placing learners and their outcomes at the center.
 - ✓ Collaboration: Building partnerships with corporates and academic institutions.

7. Technology Infrastructure

LiveTechSkills has invested in modern technology to support both training and consulting services:

- **Learning Management System (LMS)** – For delivering online courses, assignments, and assessments.
- **Cloud Labs** – Virtual labs where students practice coding, data analysis, and machine learning without hardware limitations.
- **Power BI & Tableau Servers** – For deploying and testing client dashboards.
- **Virtual Machines (VMs)** – For simulating corporate IT environments, cybersecurity exercises, and big data workloads.
- **ETL Tools** – Power Query, SQL, and Python-based ETL pipelines for project execution.

8. Key Achievements

- **1000+ alumni placed** in reputed IT firms across India.
 - **50+ corporate training programs** delivered in domains like cloud, data science, and cybersecurity.
 - **Partnerships with colleges and universities**, enabling internships and placement tie-ups.
 - **Development of in-house placement portal**, connecting trained candidates with employers.
 - **Client projects** delivered in supply chain analytics, predictive modeling, and cybersecurity audits.
 - Recognized as a trusted training partner in Maharashtra and Central India.
-

9. SWOT Analysis



1. Strengths

- Industry-relevant curriculum.
- Experienced trainers.
- Strong corporate tie-ups.
- Effective use of hybrid Agile + Waterfall methods.

2. Weaknesses

- Limited presence across India.
- Dependence on physical centers for certain services.

3. Opportunities

- Expansion into online learning platforms.
- Collaborations for certifications with reputed bodies.
- Scope to offer nationwide corporate consulting.

4. Threats

- Intense competition from larger national training brands.
 - Rapidly changing technology requires frequent course updates.
-

10. Future Plans

Looking ahead, LiveTechSkills aims to **expand nationally** by launching centers in metro cities and increasing its online presence. Plans include:

- **Partnerships with global certification bodies** like AWS, Microsoft, and Cisco.
- Launching **subscription-based online courses** for lifelong learning.
- Expanding **consulting services in data engineering, AI, and SCM analytics**.
- Building **international collaborations** for cross-border training and placements.
- Investing in **AI-driven personalized learning paths** for students.

(B) Sector Overview

1. Overview of IT Training Industry in India

1. **Evolution of IT Training:** Shift from basics like C, C++, Java to advanced fields like Data Science, AI, Cloud Computing, Cybersecurity, and Business Analytics.
 2. **Skill Demand:** India produces 1.5M engineering graduates yearly; only a fraction are industry-ready, creating a skill gap.
 3. **Market Growth:** IT training market in India expected to grow at 15–18% CAGR (2023–2030) due to digital transformation and AI adoption.
 4. **Corporate Training:** Companies like TCS, Infosys, Wipro, Cognizant invest in in-house academies; private players (e.g., LiveTechSkills) provide courses for students, graduates, and SMEs.
 5. **Reskilling Needs:** Rapid technology changes demand continuous learning in tools like Power BI, Tableau, Python, AWS.
 6. **Online Learning:** Pandemic accelerated virtual labs, LMS platforms, and remote training.
 7. **Industry-Academia Link:** Training providers integrate academic knowledge with real projects to ease student transition into professional roles.
-

2. Global SCM Analytics Market (Power BI/Tableau Use)

1. **Global SCM Analytics Market:** Expected to reach **USD 13–15 billion by 2030**, driven by E-commerce growth, globalization, and need for real-time visibility.
 2. **BI Tools in SCM:** Platforms like Power BI, Tableau, and Qlik help optimize supply chains.
 3. **Key Benefits:**
 - Real-time inventory monitoring across warehouses.
 - Demand forecasting using historical and external data.
 - Supplier performance tracking (On-Time Delivery, Lead Time, Cost Variance).
 - Logistics optimization (fuel consumption, delivery times, route efficiency).
 4. **Power BI & Tableau:**
 - Power BI: Affordable, Microsoft-integrated, easy dashboards for analysts and managers.
 - Tableau: Strong in interactive visualizations and storytelling.
 5. **Impact:** Companies can **reduce logistics costs by 10–15%** and **improve service levels by 20–25%**.
 6. **Examples:** Amazon (global inventory tracking), Flipkart & BigBasket (real-time demand-supply dashboards).
-

3. Role of IT in Supply Chain Digitization

1. **Supply Chain Evolution:** Shift from linear flow of goods to a digitally connected ecosystem.
 2. **Role of IT:** Enables real-time data exchange, end-to-end visibility, and predictive analytics.
 3. **Traditional vs Digital:** Manual logs and fragmented ERPs → Cloud ERP, IoT, blockchain, BI dashboards.
 4. **Key IT Contributions:**
 - End-to-End Visibility: Real-time updates on raw materials, inventory, and deliveries.
 - Integration Across Partners: APIs enable seamless collaboration between suppliers, manufacturers, distributors, and retailers.
 - Data-Driven Decisions: BI dashboards support demand forecasting, supplier risk management, and logistics optimization.
 - Automation: RPA streamlines tasks like order processing and invoice matching.
 - Blockchain: Ensures transparency and traceability in procurement and logistics.
 5. **Indian Industry Adoption:** FMCG, manufacturing, and e-commerce sectors digitizing supply chains to handle disruptions.
 6. **Support by IT Firms:** Companies like LiveTechSkills provide custom dashboards, predictive models, and BI tool training.
-

4. Trends in BI & Analytics (AI, IoT, Digital Twins)

1. **AI-Driven Forecasting:** Machine learning predicts demand accurately, integrating social media trends, weather, and economic indicators.
 2. **IoT (Internet of Things):** Smart sensors provide real-time data on location, temperature, and humidity, crucial for perishables and pharmaceuticals.
 3. **Digital Twins:** Virtual replicas of supply chains simulate disruptions, demand surges, or logistic bottlenecks before real-world implementation.
 4. **Self-Service Analytics:** User-friendly BI platforms let managers create dashboards without IT support.
 5. **Sustainability & ESG Analytics:** Tracks carbon footprint, energy usage, and waste, helping optimize sustainability metrics.
 6. **Global Adoption:** Companies like DHL, Unilever, and Walmart are implementing these trends; Indian firms can leverage tools like Power BI with AI and predictive analytics features.
-

5. Regulations & Compliance in Analytics (GDPR, ESG Reporting)

1. **Stricter Data Regulations:** Global rules like GDPR enforce privacy, consent, and transparency for personal data in supply chains.
2. **ESG Reporting:** Companies must disclose environmental impact, supplier diversity, and ethical practices; analytics tracks Scope 1, 2, 3 emissions, supplier certifications, and labor standards.
3. **Indian Compliance:** Digital Personal Data Protection Act (2023) governs handling of consumer and vendor data; non-compliance leads to penalties and reputational risks.
4. **Role of BI Tools:** Analytics platforms now include role-based access, encryption, audit trails, and compliance dashboards.
5. **Industry Application:** Firms like LiveTechSkills train clients to use Power BI for governed dashboards, tracking regulatory standards and ESG metrics.

6. Future Outlook of SCM + IT Analytics

1. **Predictive & Prescriptive Analytics:** Shift from reporting past events to forecasting future scenarios and recommending actions.
 2. **Real-Time AI Supply Chains:** AI autonomously reroutes shipments and adjusts inventory during disruptions.
 3. **Hyper-Personalized Demand Forecasting:** Analytics predicts demand at SKU and individual customer segment levels.
 4. **Global Data Sharing Standards:** Blockchain and consortium-led protocols enable transparency across multi-country supply chains.
 5. **Sustainability Integration:** ESG metrics and carbon footprint tracking become mandatory in supply chains.
 6. **Analytics-as-a-Service (AaaS):** SMEs adopt cloud-based, pay-as-you-go BI platforms like Power BI and Tableau.
 7. **Indian Outlook:** Initiatives like Make in India, Digital India, and National Logistics Policy promote digitization; firms like LiveTechSkills support talent training and BI implementation.
-

PROJECT WORK — SUPPLY CHAIN MANAGEMENT ANALYTICS USING POWER BI

1. Project Objective & Scope

Objective:

- Transform raw supply chain data into actionable insights using Power BI.
- Monitor procurement, inventory, order fulfillment, and logistics performance.
- Compute and track SCM KPIs like vendor performance, inventory efficiency, transport costs, and order fill rates.
- Enable interactive dashboards for operational and executive decision-making.
- Suggest process improvements to reduce costs and enhance service levels.

Scope:

- Consolidation of structured data: orders, receipts, shipments, purchase orders, vendor & product master data.
- ETL transformations using Power Query.
- Data modeling in Power BI with star schema.
- KPI calculation and validation using DAX.
- Dashboard creation for Procurement, Inventory, Logistics, and Sales/Orders.
- Pilot deployment on Power BI Service with scheduled refreshes and access control.

Business Need:

- Clients often face fragmented systems (ERP, WMS, spreadsheets), causing delayed decisions.
- Leads to overstocks, stockouts, high shipping costs, and poor supplier accountability.
- Power BI provides unified visibility for faster root-cause analysis and data-driven decision-making.

2. Data Sources (Structure & Content)

Operational / Transactional Sources:

- **ERP Order Exports (Fact_Orders):** Order details including OrderID, OrderDate, ProductID, Quantity, Price, CustomerID, Status, Fulfillment.
- **Purchase Orders / Receipts (Fact_PurchaseOrders / Fact_Receipts):** POID, VendorID, POQty, ReceivedQty, Dates, InvoiceAmount.
- **Inventory Snapshots (Fact_Inventory):** Periodic inventory by Product, Location, OnHandQty, OnHandValue.
- **Shipment Events (Fact_Shipments):** ShipmentID, OrderID, Carrier, Ship/Delivery Dates, Cost, TrackingStatus.
- **Vendor Master (Dim_Vendor):** Vendor details, lead time, country, tier.
- **Product Master (Dim_Product):** Product details, category, unit cost, reorder point, lead time.
- **Location Master (Dim_Location):** Warehouse/region info.
- **Date Dimension (Dim_Date):** Calendar table with dates, months, quarters, fiscal periods, weekdays.

Semi-structured / External Sources:

- Vendor spreadsheets or scorecards.
- Transport invoices (CSV/PDF) for logistics costs.
- Optional external signals: fuel prices, promotions, weather for demand sensing.

Data Quality Considerations:

- Ensure unique identifiers across systems; use surrogate keys if needed.
- Normalize date formats and time zones.
- Track source lineage (SourceSystem column) for auditability.
- Mask or remove PII to comply with data privacy rules.

3. ETL Pipeline (Power Query) — Steps & Best Practices

ETL Pipeline Using Power Query (M Language)

High-level Steps:

1. **Landing & Cataloguing:** Store raw CSVs/ERP extracts in a landing folder or blob; maintain a manifest with file names, dates, and schemas.
2. **Initial Load / Schema Detection:** Create queries for each source (Orders, POs, Receipts, Inventory, Shipments, Masters); set explicit data types.
3. **Cleaning & Standardization:** Trim text, standardize product/vendor codes, handle nulls, correct dates, and flag invalid values (e.g., negative quantities).
4. **Enrichment:** Merge master data with transactional tables; compute derived fields like LeadTimeDays and LineValue; normalize currency.
5. **Deduplication & Validation:** Remove duplicates, validate totals against source, and generate reconciliation tables for mismatches.
6. **Aggregation:** Pre-aggregate high-cardinality fields (daily inventory, weekly vendor totals) to enhance report performance.
7. **Load to Model:** Load cleaned and shaped tables into Power BI; for large datasets, use Azure SQL, Dataflows, or composite/direct query models.

4. Data Model in Power BI (Star Schema & Relationships)

Fact Tables:

- **Fact_Orders:** Order lines with keys (OrderID, DateKey, ProductKey, LocationKey), OrderedQty, FulfilledQty, LineValue.
- **Fact_Purchase:** POs & receipts with POID, DateKeys, VendorKey, ProductKey, POQty, ReceivedQty, InvoiceAmount.
- **Fact_Inventory:** Periodic snapshots with DateKey, ProductKey, LocationKey, OnHandQty, OnHandValue.
- **Fact_Shipments:** Shipments with ShipmentID, OrderID, Ship/Delivery DateKeys, Carrier, ShipCost.

Dimension Tables:

- **Dim_Product:** Product attributes & UnitCost.
- **Dim_Vendor:** Vendor details & lead-time standards.
- **Dim_Date:** Full calendar with fiscal markers.
- **Dim_Location:** Warehouses & regions.
- **Dim_Customer:** Optional for customer-level analysis.

Relationships & Modeling:

- Single-direction relationships from dimensions to facts; many-to-one cardinality.
- Use surrogate integer keys for performance.
- Disable auto time intelligence if using custom Dim_Date.
- Pre-aggregate tables (e.g., Agg_Daily_Inventory) for faster visuals.
- Prefer measures over calculated columns; use USERELATIONSHIP for alternate date analyses.

5. KPIs Designed (Definition, Business Purpose, Visualization)**Fact Tables:**

- **Fact_Orders:** Order lines with keys (OrderID, DateKey, ProductKey, LocationKey), OrderedQty, FulfilledQty, LineValue.
- **Fact_Purchase:** POs & receipts with POID, DateKeys, VendorKey, ProductKey, POQty, ReceivedQty, InvoiceAmount.
- **Fact_Inventory:** Periodic snapshots with DateKey, ProductKey, LocationKey, OnHandQty, OnHandValue.
- **Fact_Shipments:** Shipments with ShipmentID, OrderID, Ship/Delivery DateKeys, Carrier, ShipCost.

Dimension Tables:

- **Dim_Product:** Product attributes & UnitCost.
- **Dim_Vendor:** Vendor details & lead-time standards.
- **Dim_Date:** Full calendar with fiscal markers.
- **Dim_Location:** Warehouses & regions.
- **Dim_Customer:** Optional for customer-level analysis.

Relationships & Modeling:

- Single-direction relationships from dimensions to facts; many-to-one cardinality.
 - Use surrogate integer keys for performance.
 - Disable auto time intelligence if using custom Dim_Date.
 - Pre-aggregate tables (e.g., Agg_Daily_Inventory) for faster visuals.
 - Prefer measures over calculated columns; use USERELATIONSHIP for alternate date analyses.
-

6. Insights Extracted (Examples & Interpretation)

1. **Vendor Lead Time Variability:** Vendor V2 shows high lead time variability (5–25 days), causing stockouts.
Action: diversify vendors or increase safety stock.
2. **Inventory Concentration (Pareto):** Top 20% SKUs account for ~68% inventory value and ~62% sales.
Action: apply ABC classification; tighten controls for A SKUs, reduce C SKU stock.
3. **Logistics Cost Spike during Promotions:** Shipping costs rose 18% during promotions due to small/expedited orders.
Action: bundle shipments or incentivize consolidated orders.
4. **Low Fill Rate in Key Region:** Region East fill rate dropped to 82% vs 95% benchmark.
Action: revise reorder points, expedite critical SKUs, improve inter-warehouse transfers.
5. **Procurement Cycle Bottlenecks:** PO cycle time averages 14 days, manual approvals add 3–4 days.
Action: automate approvals and standardize invoice/receipt processes.

7. Business Impact & ROI Estimation

- **Working Capital Reduction:** Improving Inventory DOI by 15% (from 90 → 76.5 days) frees ~₹37 lakh in working capital, improving cash flow and reducing financing costs.
- **Logistics Cost Savings:** Analytics-driven route optimization reduces monthly logistics costs by 8%, saving ~₹4 lakh/month (~₹48 lakh/year).
- **Vendor Performance & Stockouts:** Reducing lead-time variability recovers ~₹2.5 lakh/month in lost sales.
- **ROI Estimation:** Implementation cost ~₹6 lakh; first-year benefits ~₹1 crore → ROI >1600%.
- **Operational Impact:** Faster root-cause analysis, data-driven vendor actions, and automated reporting reduce manual effort and save headcount hours.

8. Agile vs. Waterfall in Project Execution (Recommended Hybrid Approach)

1. **Waterfall (Initial Phase):**
 - Project initiation, scope, and architecture design.
 - Data access, compliance checks, and source agreements
2. **Agile (Iterative Phase):**
 - Dashboard development, KPI refinement, and stakeholder feedback.
 - Sprint demos, rapid changes, and iterative improvements.
3. **Sample Sprint Plan (6×1 week or 3×2 weeks):**
 - Sprint 0: Data access, profiling, finalize scope.
 - Sprint 1–3: ETL & dashboard prototypes (Orders, Inventory, Shipments).
 - Sprint 4: KPI validation & DAX optimization.
 - Sprint 5: UAT, role-based access, documentation, training.
 - Sprint 6: Deployment to Power BI Service, scheduled refresh, handover.

Roles & Artifacts

- **Product Owner / Client Sponsor** — defines & prioritizes KPIs.
- **BI Developer / Data Engineer** — ETL, model, DAX measures.
- **Data Analyst / Business Analyst** — validates metrics and interprets insights.
- **IT / DevOps** — data access, Power BI Service workspace, scheduled refresh.
- **Artifacts:** Project charter, data dictionary, KPI definitions, sprint backlog, UAT sign-off.

Advantages of hybrid

- Faster visible value (Agile) while reducing risk (Waterfall for data governance).
 - Close collaboration with stakeholders ensures dashboards reflect realistic KPIs.
 - Enables incremental ROI — early wins (e.g., top 10 vendor remediation) can be realized quickly.
-

2.(A) Actual Work Done

Weeks	Learning
Week 1	Orientation & Requirement Gathering
Week 2	Data Collection & Cleaning
Week 3	ETL & Data Modeling
Week 4	KPI Design in DAX
Week 5	Dashboard Development
Week 6	Client Feedback
Week 7	Finalization

Week-wise Work Done

Week 1 – Orientation & Requirement Gathering

- Attended orientation session to understand the company, project scope, and objectives.
 - Joined client introduction meeting, learned about supply chain issues like late deliveries and stockouts.
 - Studied the existing supply chain process (procurement, inventory, logistics, sales flow).
 - Collected raw reports and data shared by the client.
 - Discussed project requirements with mentor and listed major goals.
 - Prepared notes on supply chain challenges and expected dashboard outputs
-

Week 2 – Data Collection & Cleaning

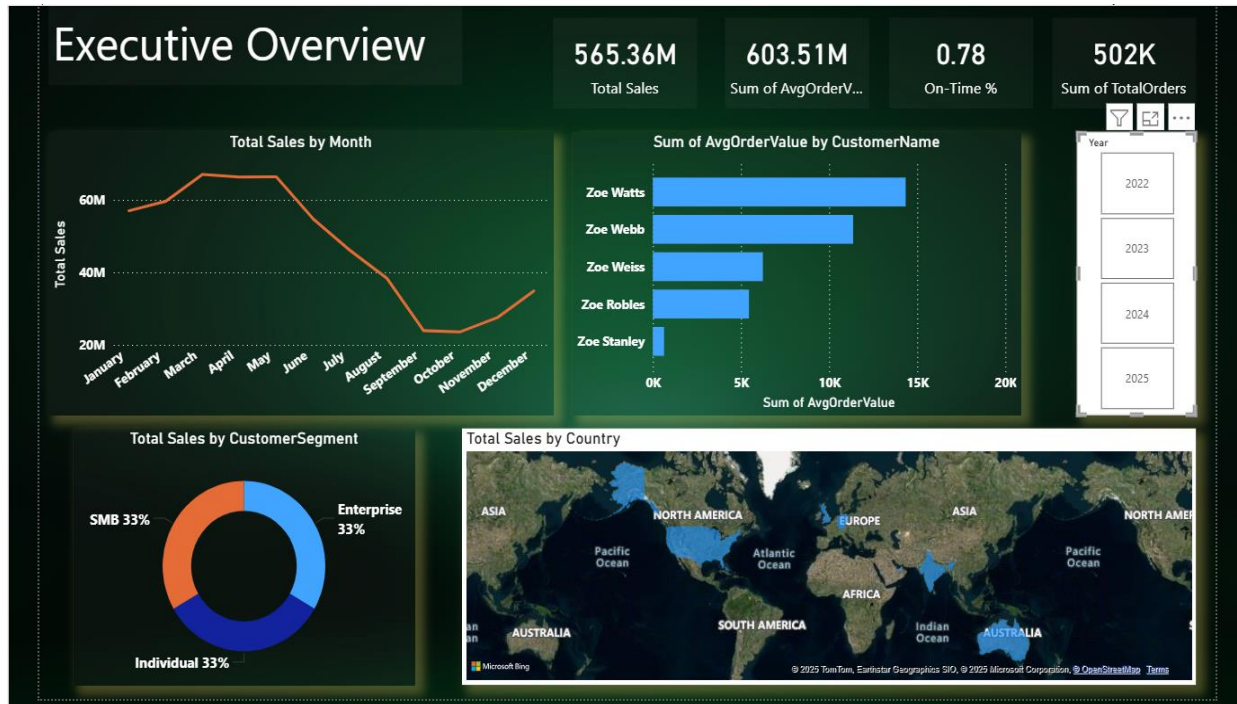
- Gathered datasets from Excel, SQL, and ERP exports.
 - Checked missing values, duplicate entries, and inconsistent formats.
 - Removed null values and corrected wrong entries.
 - Standardized supplier codes and product names to avoid mismatches.
 - Handled outliers in delivery time and cost data for accuracy.
 - verified cleaned data with original reports.
-

Week 3 – ETL & Data Modeling

- Loaded datasets into Power BI using Power Query.
 - Applied transformations like merging, splitting, and unpivoting columns.
 - Changed data types and created calculated columns.
 - Designed dimension tables (Supplier, Product, Region, Time).
 - Created fact table with Orders and Transactions.
 - Built relationships between tables to form a star schema model.
-

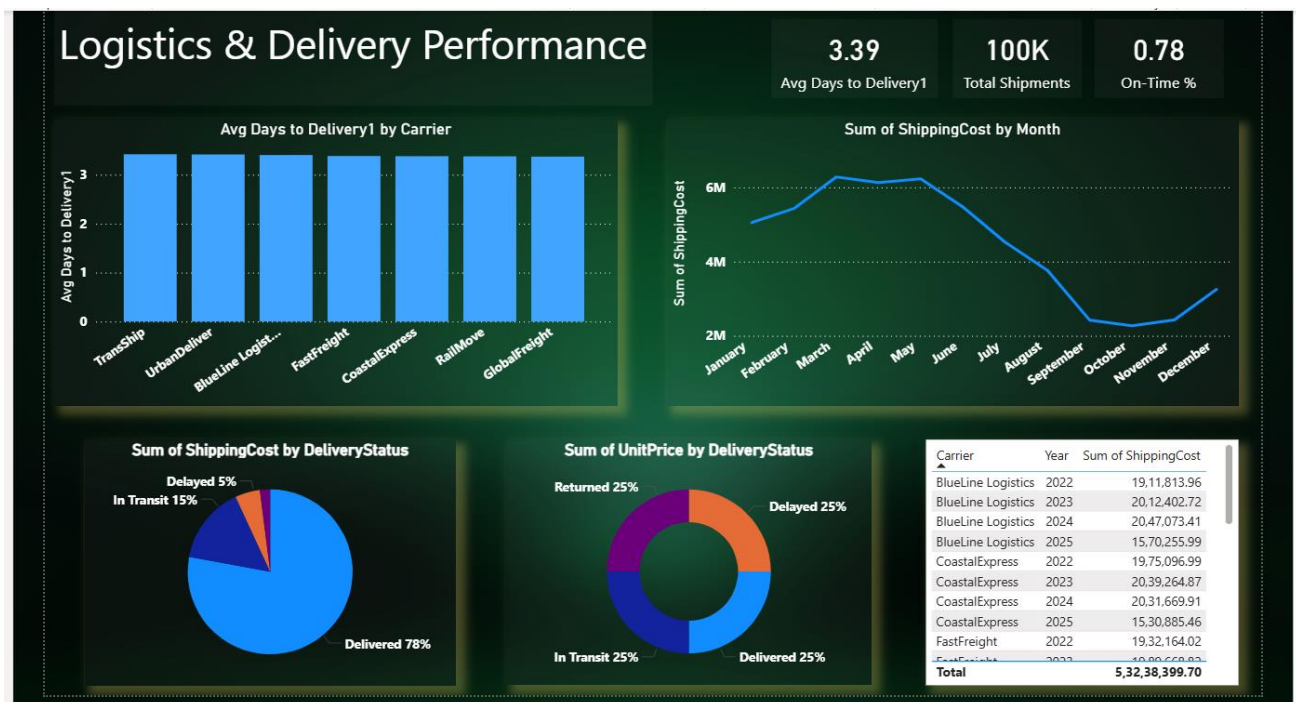
Week 4 – KPI Design in DAX

- Learned basics of DAX formulas and syntax.
- Created a measure for *Total Orders*.
- Built Average Lead Time calculation.
- Designed Inventory Turnover Ratio.
- Added Supplier Performance Score.
- Cross-checked values with client-provided numbers for validation.



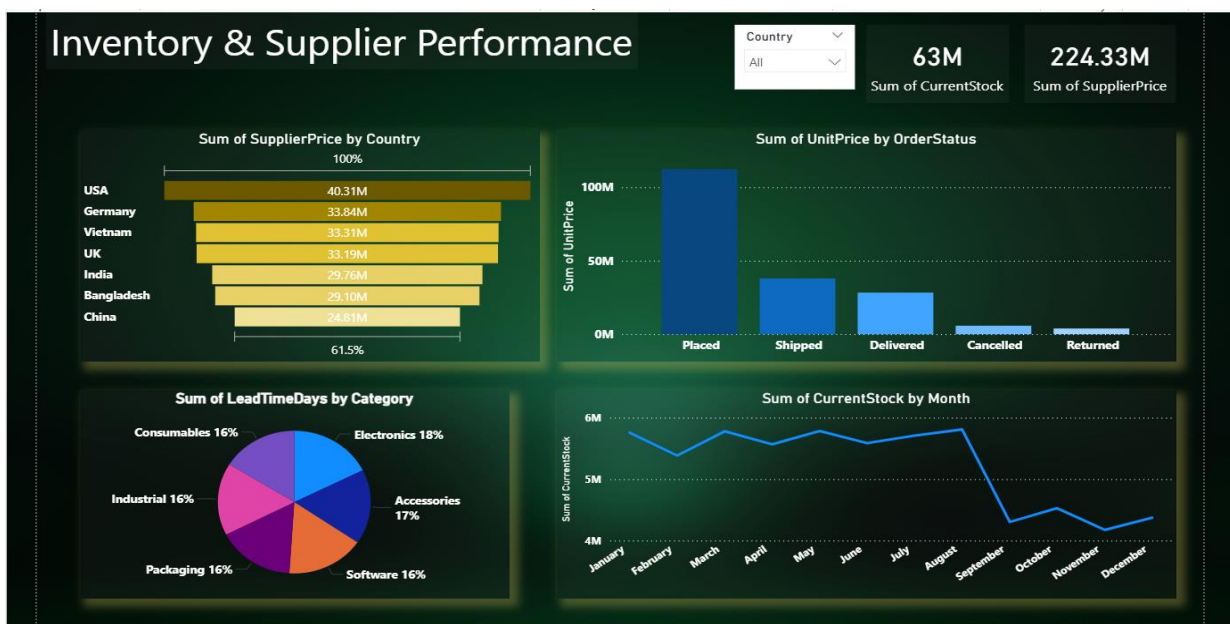
Week 5 – Dashboard Development

- Built **interactive dashboards** in Power BI using effective visuals (bar charts, line graphs, maps, and cards).
- Added **slicers** for time, region, and supplier selection.
- Implemented **drill-through pages** for deeper insights into supplier performance and logistics.
- Focused on clear, business-friendly design with easy interpretation for non-technical users



Week 6 – Client Feedback

- Presented the initial version of dashboards to the client.
- Collected feedback regarding missing insights and requested KPIs.
- Iterated on reports by adding new metrics (e.g., Cost per Order, Delayed Shipment Percentage).
- Enhanced visual storytelling based on client suggestions for decision-making support.



Week 7 – Finalization

- Finalized all dashboards with updated measures and design improvements.
- Conducted testing to ensure reports were accurate, interactive, and optimized.
- Prepared **project documentation** (data dictionary, KPI definitions, methodology notes).
- Delivered final presentation to client and stakeholders highlighting supply chain insights and actionable recommendations.

Top 25 Suppliers Name And Details

Category	Industrial	Packaging	Software			Total	
SupplierName	Sum of ReliabilityScore	Sum of LeadTimeDays	Sum of ReliabilityScore	Sum of LeadTimeDays	Sum of ReliabilityScore	Sum of LeadTimeDays	Sum of ReliabilityScore
Yang, Wilson and Zimmerman	18.82	969	19.87	1195	25.50	6181	133.83
Young Ltd	15.16	352	11.41	418	13.31	2571	86.93
Young, Contreras and Marshall	14.01	614	13.35	690	15.41	4058	90.92
Young-Allen	15.83	1006	20.43	463	9.95	4648	95.32
Young-Martinez	17.14	68	19.66	98	16.23	500	109.59
Young-Molina	15.45	1162	22.50	935	18.87	5448	107.72
Young-Walter	14.11	86	18.78	94	14.94	484	83.61
Yu-Brooks	17.03	753	15.44	611	12.81	4652	97.50
Zhang PLC	13.52	338	10.89	429	13.67	2374	77.50
Zhang, Brooks and Miles	17.73	401	12.50	826	24.79	3879	115.33
Total	158.80	5749	164.83	5759	165.48	34795	998.25

B. Findings & Suggestions

Introduction

The analysis of the Supply Chain Management (SCM) data using Power BI revealed several patterns, bottlenecks, and opportunities for improvement. Findings are based on dashboards covering **Procurement, Inventory, Logistics, and Sales**. The insights highlight operational inefficiencies, cost leakages, and risks that, if not addressed, could impact profitability and customer satisfaction. Each finding is discussed in detail, followed by data-driven suggestions to improve overall supply chain performance.

1. Finding: Vendors A & B Caused 40% of Delivery Delays

- **Observation:** The Vendor Performance Dashboard shows that Vendors A and B accounted for nearly 40% of all late deliveries. The On-Time Delivery (OTD) KPI for Vendor A was 62% and for Vendor B 68%, both significantly lower than the company benchmark of 90%. Vendor lead times were also inconsistent, with some shipments arriving 10–12 days later than the agreed standard.
- **Business Impact:** These delays caused downstream disruptions — stockouts at warehouses, increased emergency procurement, and poor order fulfillment rates for customers. Frequent vendor unreliability directly impacts customer satisfaction and erodes trust with end-users.

Suggestion:

- Develop Vendor Scorecards in Power BI with KPIs such as On-Time Delivery %, Average Lead Time, and Defect Rate.
 - Use these scorecards for quarterly vendor evaluations and negotiations.
 - Implement a Preferred Vendor Program, reducing dependency on underperforming suppliers by shifting volumes to more reliable vendors.
 - For Vendors A & B, initiate corrective actions: contract penalties for repeated delays or collaborative improvement programs.
-

2. Finding: Logistics Costs Peaked in Q2 Due to Fuel Hikes

- **Observation:** The Logistics Dashboard revealed that transportation costs surged by 18% in Q2 compared to Q1. The spike correlated with external factors, primarily rising fuel prices. Additionally, a higher proportion of shipments were expedited due to delayed procurement, further inflating logistics costs.
- **Business Impact:** Logistics costs reached 12% of sales revenue in Q2 versus the benchmark of 8%. This reduced profit margins and increased the cost-to-serve, especially in regions that relied on long-distance transport.

Suggestion:

- Optimize transportation routes using route analytics to reduce empty miles and unnecessary detours.
- Consolidate smaller shipments into full-truck loads wherever possible to achieve economies of scale.
- Establish long-term contracts with logistics providers to mitigate the effect of fluctuating fuel prices.
- Incorporate external data (fuel indexes, toll data) into Power BI dashboards to predict logistics cost surges in advance and prepare contingency plans.

3. Finding: Inventory Holding Costs Increased Due to Poor Demand Forecasting

- **Observation:** The Inventory Dashboard showed that average inventory holding costs increased by 22% in the last financial year. Days of Inventory (DOI) rose from 75 days to 92 days. The excess stock was primarily due to inaccurate demand forecasting, where demand for certain SKUs was overestimated. On the other hand, some fast-moving items suffered stockouts due to underestimation.
- **Business Impact:** Higher holding costs tied up working capital, reduced liquidity, and increased warehouse rental expenses. Overstocked items risked obsolescence, while stockouts damaged customer trust and caused lost sales opportunities.

Suggestion:

- Introduce Predictive Demand Forecasting models (using Python, R, or Azure ML) and integrate the outputs into Power BI.
- Build dashboards that compare forecast vs. actual demand to track forecasting accuracy.
- Classify products using ABC analysis: allocate tighter forecasting accuracy for high-value “A” products and adopt simpler methods for “C” products.
- Align procurement schedules with refined demand forecasts to reduce excess stock.

4. Finding: Inventory Imbalances Across Warehouses

- **Observation:** The dashboard highlighted imbalances in stock distribution across warehouses. While some warehouses maintained over 120 days of stock for certain products, others faced frequent stockouts. Transfer orders were initiated late, causing customer delivery delays in understocked regions.
- **Business Impact:** Unequal distribution increased transfer costs and led to poor service levels in high-demand regions. This imbalance also contributed to higher inventory carrying costs company-wide.

Suggestion:

- Use multi-location inventory optimization models to balance stock across warehouses.
- Automate stock reallocation triggers in Power BI: when stock in one warehouse exceeds 90 days and another falls below 30 days, generate transfer recommendations.
- Implement dynamic safety stock policies based on regional demand variability.

5. Finding: Procurement Cycle Time is Longer Than Industry Benchmark

- **Observation:** Analysis of procurement processes showed that the average cycle time from purchase order creation to goods receipt was 14 days, compared to an industry benchmark of 9–10 days. A significant portion of this delay (3–4 days) was due to manual approvals and paper-based processes.
- **Business Impact:** Extended procurement cycles delay production schedules, increase the need for expedited logistics, and reduce supply chain agility.

Suggestion:

- Automate procurement workflows using ERP integration and digital approval systems.
- Introduce thresholds: for small-value, low-risk purchases, bypass multi-level approvals.
- Use Power BI to monitor procurement cycle times and highlight bottlenecks.
- Establish SLAs with suppliers to reduce approval-to-delivery delays.

6. Finding: Lack of Automated Data Refreshes in Reporting

- **Observation:** During the project, it was observed that many teams relied on manual Excel-based reports, updated weekly or monthly. This delayed visibility into critical KPIs such as fill rates, vendor OTD %, and logistics costs.
- **Business Impact:** Decision-makers lacked real-time insights, leading to reactive firefighting instead of proactive planning.

Suggestion:

- Configure automated data refresh schedules in Power BI Service (daily or hourly depending on business need).
- Establish a single source of truth by centralizing all data in Power BI datasets instead of multiple Excel sheets.
- Train managers to use self-service BI features for instant analysis.
-

Finding	Business Impact	Suggestion
Vendors A & B caused 40% delays	Stockouts, poor OTD	Vendor scorecards, corrective action, preferred vendors
Logistics costs peaked in Q2	Profit margin erosion	Route optimization, shipment consolidation, long-term contracts
Inventory holding costs increased	Higher working capital, obsolescence	Predictive demand forecasting, ABC analysis
Inventory imbalances across warehouses	Poor service levels, transfer costs	Multi-location optimization, automated transfer triggers
Procurement cycle longer than benchmark	Production delays, expedited costs	ERP integration, digital approvals, SLA enforcement
Lack of automated reporting	Reactive decisions	Automated refresh in Power BI, central data model

(C) Conclusions

Power BI Improved Transparency

The Summer Internship Project on Supply Chain Management Analytics Using Power BI has significantly demonstrated how Business Intelligence tools improve organizational transparency and decision-making. Before the project, most supply chain reporting was fragmented across spreadsheets, departmental reports, and email-based updates. This led to delays in information flow and inconsistent data interpretation.

By integrating supplier, procurement, inventory, logistics, and sales data into Power BI dashboards, decision-makers were able to view real-time KPIs such as Order Fill Rate, Vendor Lead Time, Inventory Turnover Ratio, and Logistics Cost %. The visualizations enabled stakeholders to drill down from aggregated company-wide metrics to granular supplier- or SKU-level insights. This single source of truth reduced miscommunication between departments and empowered leaders to take timely corrective actions.

For example, vendor performance dashboards quickly highlighted that Vendors A and B were responsible for 40% of all late deliveries. Instead of manually investigating multiple Excel reports, the supply chain team could now access these insights instantly, enabling faster vendor negotiations and contract reviews. Similarly, inventory dashboards showed imbalances across warehouses, helping managers optimize transfers proactively rather than reacting after stockouts.

Thus, the implementation of Power BI improved data transparency, accountability, and responsiveness, aligning the client's supply chain operations with modern, data-driven business practices.

Role of Agile and Waterfall in Project Execution

The project followed a hybrid methodology, combining elements of both Agile and Waterfall approaches.

- **Waterfall Approach:** The project began with traditional waterfall steps — requirement gathering, data collection, and ETL pipeline development. These stages required structured documentation and sequential execution to ensure data integrity and reliability before analytics could begin. For example, cleaning and merging supplier and logistics datasets was a prerequisite to creating a unified data model.
- **Agile Approach:** Once the data foundation was set, the dashboard design and KPI development were executed in iterative sprints, following Agile principles. Each week, a set of dashboards (procurement, inventory, logistics) was built and presented to stakeholders for feedback. Rapid iterations allowed for incremental improvements, ensuring dashboards reflected real business needs rather than static assumptions.

This combination proved effective: **Waterfall provided structure** for critical backend tasks, while **Agile offered flexibility** for user-facing dashboards and insights. The hybrid model balanced documentation with adaptability, a key requirement for analytics projects in dynamic business environments.

Key Learnings from the Internship

The internship offered rich learning experiences across both technical and managerial dimensions.

Technical Learnings

1. **ETL Pipeline Creation in Power Query** – Hands-on experience with data cleaning, merging, and transformation improved understanding of how raw business data can be shaped for reporting.
2. **Data Modeling in Power BI** – Designing fact and dimension tables, managing relationships, and ensuring performance optimization were critical technical skills developed.
3. **DAX (Data Analysis Expressions)** – The project required the use of advanced DAX formulas to calculate KPIs such as Inventory Turnover Ratio, Procurement Efficiency, and Logistics Cost %. Writing and debugging these formulas sharpened analytical thinking.
4. **Dashboard Design** – Practical exposure to building user-friendly dashboards reinforced design principles such as clarity, consistency, and actionable insights.
5. **Integration with Business Processes** – The project showed how BI tools are not standalone but must be aligned with operational processes like procurement cycles, vendor management, and logistics optimization.

Managerial Learnings

1. **Stakeholder Communication** – Weekly reviews with mentors and clients taught the importance of presenting data in a simple, decision-oriented manner.
 2. **Time Management** – Balancing data preparation, technical implementation, and documentation within the 6-week internship period improved efficiency and prioritization skills.
 3. **Agility in Problem Solving** – Encountering data inconsistencies and last-minute scope changes required adaptability and quick problem resolution.
 4. **Collaboration in Cross-Functional Teams** – Working with professionals from procurement, inventory, and logistics departments highlighted the importance of understanding diverse perspectives in business projects.
 5. **Documentation & Reporting** – Preparing structured reports and maintaining weekly work logs emphasized the role of thorough documentation in academic as well as corporate environments.
-

Benefits to Company & Client

The project created tangible and intangible benefits for both the internship company (LiveTechSkills) and the client organization whose supply chain data was analyzed.

For the Company (LiveTechSkills)

- Strengthened its portfolio of analytics projects, especially in the supply chain vertical, which is increasingly in demand among clients.
- Demonstrated capability in combining Agile and Waterfall methodologies, which can be used as a delivery model in future consulting engagements.
- Enhanced credibility with potential clients by showcasing a live, impactful use case of Power BI in optimizing supply chains.
- Provided learning opportunities for internal mentors and trainers, enriching their expertise in real-world BI projects.

For the Client Organization

- Gained real-time visibility into vendor, procurement, and logistics performance.
- Identified cost-saving opportunities, such as optimizing transportation routes and reducing inventory holding costs.
- Built a foundation for predictive analytics, enabling future adoption of AI-driven demand forecasting.
- Improved decision-making speed by replacing static Excel reports with dynamic, automated dashboards.
- Enhanced vendor accountability through transparent performance tracking.

Final Reflection

The internship underscored the transformative role of analytics in modern business. By applying Power BI to supply chain data, the project demonstrated how organizations can move from descriptive reporting to prescriptive decision-making. The hybrid Agile–Waterfall approach ensured both structural rigor and iterative flexibility.

On a personal level, the project provided valuable exposure to real-world data challenges, advanced BI techniques, and cross-functional teamwork. On an organizational level, it delivered actionable insights that can directly improve operational efficiency and profitability.

In conclusion, the SIP was a mutually beneficial engagement that combined academic learning with corporate impact, bridging the gap between theory and practice.

Bibliography

1. Books & Academic References

- Chopra, S., & Meindl, P. (2021). Supply Chain Management: Strategy, Planning, and Operation. Pearson Education.
- Hugos, M. H. (2018). Essentials of Supply Chain Management. Wiley.

2. Industry Reports

- NASSCOM (2023). India IT Industry Report.
- Gartner (2024). Global Supply Chain Analytics Market Outlook.

3. Websites

- Microsoft Power BI Documentation. Retrieved from <https://learn.microsoft.com/power-bi>
- LiveTechSkills Official Website. Retrieved from <https://www.livetechskills.com>

4. Research Articles / Journals

- Articles on SCM analytics, data visualization, or BI adoption (Google Scholar or ResearchGate).
- Case studies on supply chain analytics from Harvard Business Review or Elsevier Journals.

5. Other References

- Lecture notes and guidance provided by faculty.
 - Company documents and raw datasets shared during internship.
-

Performance Letter



KDK COLLEGE OF ENGINEERING
(An Autonomous Institute)
DEPARTMENT OF MANAGEMENT STUDIES

**PERFORMANCE SHEET FROM ORGANISATION FOR
SIP**

STUDENT (Mentee)

(To be filled by the Organisation Supervisor at the end of SIP)

Name of the Student: Vaishnavi . Amit . Ojha.

Organisation Name: Livetechnskills

Location/Work Place Address: New Nandorvan, MB Tower

Organisation Supervisor (Guide) Name: Gaurav . Bansod

Contact No: 9156511207 Email ID: gauravb@edvercity.com

Dear Sir/Madam,

Please rate the intern on a scale of 1-10 (1 being poor and 10 being excellent) for the below mentioned parameters.

Sr. No	Parameter of your Evaluation out of 10	Your Rating out of 10 for Mentee
1	Knowledge of Subject Area	10
2	Punctuality	9
3	Attitude	10
4	Communication Skill – Written	10
5	Communication Skill – Oral	10
6	Accomplishment of Assigned Tasks	9
7	Confidence Level	10
8	General Awareness	10
9	Grasping Ability	9
10	Overall Personality	10

Your Suggestion for Future Improvement Always being polite and
good.


FOR LIVETECHSKILLS
Signature & Seal of the Organisation Guide
PARTNER